

Signal Processing

Exercise 3

Solution

SoSe26

Linearity

System 1.1

Step (1): $\mathfrak{S}(\alpha x_1(t) + \beta x_2(t)) = \frac{d}{dt} [\alpha x_1(t) + \beta x_2(t)]$

Step (2) Expected output if linear: $y_r(t) = \alpha \frac{d}{dt} [x_1(t)] + \beta \frac{d}{dt} [x_2(t)]$

Step (3): The equations of step (1) and step (2) are equal. Therefore, the system is linear.

System 1.2

Step (1): $\mathfrak{S}(\alpha x_1(t) + \beta x_2(t)) = \frac{d}{dt} [\alpha x_1(t) + \beta x_2(t)] + 2$

Step (2) Expected output if linear:

$$y_r(t) = \frac{d}{dt} [\alpha \cdot x_1(t)] + 2 + \frac{d}{dt} [\beta \cdot x_2(t)] + 2 = \alpha \frac{d}{dt} [x_1(t)] + 2 + \beta \frac{d}{dt} [x_2(t)] + 2$$

Step (3): The equations of step (1) and step (2) are not equal. Therefore, the system is not linear.

System 1.3

Step (1): $\mathfrak{S}(\alpha x_1(t) + \beta x_2(t)) = \int_{-\infty}^{t-1} [\alpha x_1(\tau) + \beta x_2(\tau)] d\tau$

Step (2) Expected output if linear: $y_r(t) = \alpha \int_{-\infty}^{t-1} x_1(\tau) d\tau + \beta \int_{-\infty}^{t-1} x_2(\tau) d\tau$

Step (3): The equations of step (1) and step (2) are equal. Therefore, the system is linear.

System 1.4

Step (1): $\mathfrak{S}(\alpha x_1(t) + \beta x_2(t)) = [\alpha x_1(t) + \beta x_2(t)]^2$

Step (2): Expected output if linear: $y_r(t) = \alpha (x_1(t))^2 + \beta (x_2(t))^2$

Step (3): The equations of step (1) and step (2) are not equal. Therefore, the system is not linear.

System 1.5

Step (1): $\mathfrak{S}(\alpha x_1(t) + \beta x_2(t)) = [\alpha x_1(t) + \beta x_2(t)] \cdot \sin(\omega_0 t)$

Step (2) Expected output if linear: $y_r(t) = \alpha x_1(t) \cdot \sin(\omega_0 t) + \beta x_2(t) \cdot \sin(\omega_0 t)$

Step (3): The equations of step (1) and step (2) are equal. Therefore, the system is linear.

Time-invariance

System 1.1

Step (1): $\mathfrak{S}(x(t-h)) = \frac{d}{dt} [x(t-h)]$

Step (2): $y_r(t-h) = \frac{d}{d(t-h)} [x(t-h)] = \frac{d}{dt} [x(t-h)]$

Step (3): The equations of step (1) and step (2) are equal. Therefore, the system time-invariant.

System 1.2

Step (1): $\mathfrak{S}(x(t-h)) = \frac{d}{dt} [x(t-h)] + 2$

Step (2): $y_r(t-h) = \frac{d}{d(t-h)} [x(t-h)] + 2 = \frac{d}{dt} [x(t-h)] + 2$

Step (3): The equations of step (1) and step (2) are equal. Therefore, the system is time-invariant.

System 1.3

Step (1): $\mathfrak{S}(x(t-h)) = \int_{-\infty}^{t-1} x(\tau-h) d\tau$

Step (2): $y_r(t-h) = \int_{-\infty}^{t-h-1} x(\tau-h) d(\tau-h)$

Step (3): We use our equation of step (1)

$$\int_{-\infty}^{t-1} x(\tau-h) d\tau$$

to show that the system is time-invariant and introduce a new variable $v = \tau - h$. Since we would like to integrate over v , the upper and lower bound become

$$v = \tau - h = t - 1 - h \text{ (upper bound) and}$$

$$v = \tau - h = -\infty \text{ (lower bound), respectively.}$$

This results in

$$\int_{-\infty}^{t-1-h} x(v) dv,$$

which is the same equation as we received in step (2). Thus, the system is time-invariant.

System 1.4

Step (1): $\mathfrak{S}(x(t-h)) = x^2(t-h)$

Step (2): $y_r(t-h) = x^2(t-h)$

Step (3): The equations of step (1) and step (2) are equal. Therefore, the system is time-invariant.

System 1.5

Step (1): $\mathfrak{S}(x(t-h)) = x(t-h) \cdot \sin(\omega_0 \cdot t)$

Step (2): $y_r(t-h) = x(t-h) \cdot \sin(\omega_0(t-h))$

Step (3): The equations of step (1) and step (2) are not equal. Therefore, the system is time-variant.

Causality/Causation

Since all systems only depend on current or past values, all systems are causal!

Watch out! If a system were to be described by

$$\mathfrak{S}(x(t)) = x(2t),$$

the system is not causal. However, if we provide certain intervals, you might have to differentiate between the intervals $] - \infty, 0]$, for which the system shows causal behavior, and $]0, \infty[$, for which the system shows non-causal behavior.